

Name: File

Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12

HSC ASSESSMENT TASK 3

JUNE 2011

MATHEMATICS Extension 1

Time Allowed: 70 minutes

Instructions:

- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a **new page**.
- Standard integrals can be found on the last page.

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/10	/10	/10	/10	/10	/10	/60

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1 (10 marks)**Marks**

- a) Differentiate $e^{\sin x}$ 1
- b) Find $\frac{d}{dx}(3^x)$ 1
- c) Find the exact value of $\tan(\sin^{-1}\frac{1}{5})$ 1
- d) Solve $\sin 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$ 3
- e) Find i) $\int \frac{1}{\sqrt{9-x^2}} dx$ 1
- ii) $\int \sin\left(\frac{\pi}{4} - x\right) dx$ 2
- iii) $\int 3xe^{x^2} dx$ 1

Question 2 (10 marks) Start a new page

- a) If $\int_0^2 \frac{dx}{x^2+4} = k$ find k 2
- b) i) Differentiate $f(x) = \tan^{-1}x + \tan^{-1}\frac{1}{x}$ if $x \neq 0$ 2
- ii) Hence evaluate $\int_1^2 f(x) dx$ 1
- c) i) Find $\frac{d}{dx}(x \ln x)$ 2
- ii) Hence show that $\int_e^{e^2} \frac{1+\ln x}{x \ln x} dx = 1 + \ln 2$ 3

Question 3 (10 marks) Start a new page**Marks**

- a) Find $\int_0^{\frac{\pi}{2}} \cos^2 4x \, dx$ 3
- b) Sketch $y = 2\sin^{-1}(x - 1)$ and state its domain and range. 3
- c) i) Write $y = \cos x - \sqrt{3} \sin x$ in the form $y = A\cos(x + B)$, where B is acute. 2
- ii) Hence, sketch $y = \cos x - \sqrt{3} \sin x$ for $0 \leq x \leq 2\pi$. Find and label clearly, the points where the curve cuts the x and y axes. 2

Question 4 (10 marks) Start a new page

- a) An area of 1 square unit is bounded by the curve $y = \frac{1}{x}$, the x axis and the lines $x = e$ and $x = k$. Find k if $k > e$. 2
- b) i) Show that $f(x) = \frac{1}{1+x^2}$ is an even function and then, without the use of calculus, make a neat sketch of $f(x) = \frac{1}{1+x^2}$. 2
- ii) What is the largest domain, containing $x = -1$, for which $f(x)$ has an inverse function $f^{-1}(x)$? 1
- iii) Find $f^{-1}(x)$ 2
- iv) Sketch $y = f^{-1}(x)$ 1
- c) Find the general solution for $\sin x = \frac{1}{2}$ 2

Question 5 (10 marks) Start a new page**Marks**

- a) i) Show that $x^2 + 6x + 10 = (x + 3)^2 + 1$ 1
- ii) Hence find $\int \frac{1}{x^2 + 6x + 10} dx$ 1
- b) i) Differentiate $x \tan^{-1} x$ 1
- ii) Hence find $\int \tan^{-1} x dx$ 2
- c) An object, removed from a freezer at -5°C , is placed in a room where the temperature is kept at a constant 15°C . Thereafter, its temperature $T^\circ\text{C}$, is changing so that after t minutes $\frac{dT}{dt} = k(15 - T)$, where k is a constant
- i) show that $T = 15 - Ae^{-kt}$ satisfies this differential equation 1
- ii) Find the value of A 1
- iii) If initially, the temperature was increasing at 5°C per min, find the value of k . 1
- iv) Find the temperature of the object, 5 mins after it was placed in the room (correct to 2 dec. pl.) 1
- v) Find to, the nearest second, the time taken for the temperature of the object to rise to 0°C . 1

Question 6 (10 marks) Start a new page

- a) A square metal plate has sides x cm and an area of A cm². It is expanding, so that the sides are increasing at 0.08 cm/min. Find the rate at which the area is increasing, when the sides are 7cm long. 3
- b) i) Sketch $y = 1 - \tan x$, in the domain $0 \leq x \leq \frac{\pi}{4}$, without the use of calculus. 2
- ii) Prove the area of the region enclosed by the above curve, the x axis and the y axis is $\frac{\pi - \ln 4}{4}$ square units. 2
- iii) The region in part ii) is rotated around the x axis. Find the volume of the solid formed. 3

Question 1

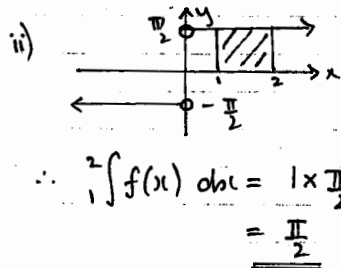
- a) $\frac{d}{dx}(e^{\sin x}) = \cos x \cdot e^{\sin x}$
- b) $\frac{d}{dx}(3^x) = \ln 3 \cdot 3^x$
- c) Let $u = \sin^{-1} \frac{1}{5}$
 $\sin u = \frac{1}{5}$

 $\therefore \tan u = \tan(\sin^{-1} \frac{1}{5}) = \frac{1}{\sqrt{24}}$
- d) $\sin 2\theta = \cos \theta$
 $2\sin \theta \cos \theta - \cos \theta = 0$
 $\cos \theta (2\sin \theta - 1) = 0$
 $\cos \theta = 0 \quad \sin \theta = \frac{1}{2}$
 $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$
- e) i) $\int \frac{1}{\sqrt{9-x^2}} dx = \sin^{-1} \frac{x}{3} + c$
- ii) $\int \sin(\frac{\pi}{4} - x) dx$
 $= \cos(\frac{\pi}{4} - x) + c$
- iii) $\int 3x \cdot e^{x^2} dx = \frac{3x e^{x^2}}{2x} + c$
 $= \frac{3e^{x^2}}{2} + c$

Question 2

- a) $\int_0^2 \frac{dx}{x^2+4} = \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2$
 $= \frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0)$
 $k = \frac{\pi}{8}$

b) i) $f(x) = \tan^{-1} x + \tan^{-1}(x^{-1})$
 $f'(x) = \frac{1}{1+x^2} + \frac{-x^{-2}}{1+x^{-2}}$
 $= \frac{1}{1+x^2} - \left[\frac{1}{x^2} \cdot \frac{x^2}{x^2+1} \right]$
 $= \frac{1}{1+x^2} - \frac{1}{x^2+1}$
 $f'(x) = 0 \text{ if } x \neq 0$



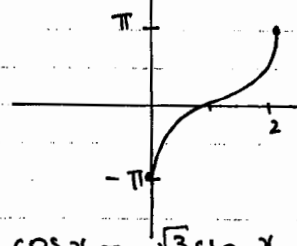
c) i) $u = x \quad v = \ln x$
 $u' = 1 \quad v' = \frac{1}{x}$
 $\frac{d}{dx}(x \ln x) = \ln x + 1$

ii) $\int \frac{e^x}{x \ln x} dx = \left[\ln(x \ln x) \right]_e^2$
 $= \ln(e^2 \ln e^2) - \ln(e \ln e)$
 $= \ln(e^2 \cdot 2 \ln e) - 1$
 $= \ln 2e^2 - 1$
 $= \ln 2 + 2 \ln e - 1$
 $= \ln 2 + 1$

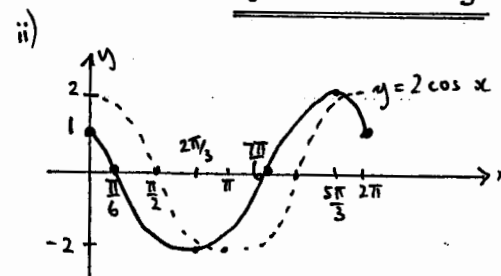
Question 3

a) $\sin \theta \cos^2 \theta = \frac{1}{2} [\cos 2\theta + 1]$
 $\therefore \int_0^{\pi/2} \frac{1}{2} (\cos 2\theta + 1) d\theta$
 $= \frac{1}{2} \left[\frac{1}{8} \sin 8x + x \right]_0^{\pi/2} = \frac{1}{2} \left[\frac{1}{8} \sin \pi + \frac{\pi}{2} \right]$
 $= \frac{\pi}{4}$

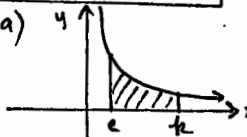
b) $y = 2 \sin^{-1}(x-1)$
 $\frac{y}{2} = \sin^{-1}(x-1)$
 $-\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}$
 $\therefore R: -\pi \leq y \leq \pi$
 $-1 \leq x-1 \leq 1$
 $D: 0 \leq x \leq 2$



c) $y = \cos x - \sqrt{3} \sin x$
 $A = \sqrt{1+3} = 2$
 $i) 2 [\cos x \cos \beta - \sin x \sin \beta]$
 $2 \left[\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x \right]$
 $\cos B = \frac{1}{2} \quad \sin B = \frac{\sqrt{3}}{2}$
 $\therefore B = \frac{\pi}{3}$
 $\therefore y = \cos x - \sqrt{3} \sin x$
 can be written $y = 2 \cos(x + \frac{\pi}{3})$

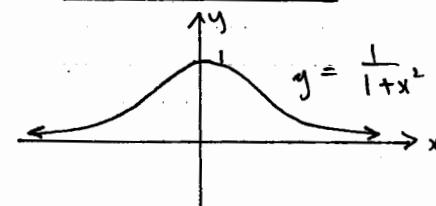


Question 4

a) 
 $1 = \int_e^k \frac{1}{x} dx$
 $1 = [\ln x]_e^k$

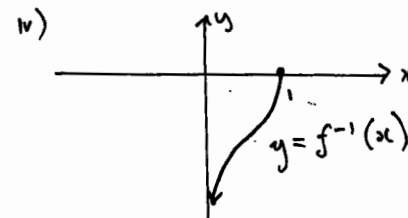
$1 = \ln k - \ln e$
 $\therefore \log_e k = 2$
 $\therefore e^2 = k$

b) i) $f(x) = \frac{1}{1+x^2}$
 $f(-x) = \frac{1}{1+(-x)^2} = \frac{1}{1+x^2}$
 $\therefore f(x) = f(-x) \text{ even}$



ii) $D: x \leq 0$

iii) $x = \frac{1}{1+y^2}$
 $\frac{1}{x} = 1+y^2$
 $\frac{1}{x} - 1 = y^2$
 $\therefore f^{-1}(x) = -\sqrt{\frac{1}{x} - 1}$



c) $\sin x = \frac{1}{2}$
 $x = n\pi + (-1)^n \sin^{-1} \frac{1}{2}$
 $= n\pi + (-1)^n \frac{\pi}{6}$
 where n is an integer

Question 5

a) i) RHS = $(x+3)^2 + 1$
 $= x^2 + 6x + 9 + 1$
 $= x^2 + 6x + 10$
 $= \text{LHS}$

ii) $\int \frac{1}{x^2 + 6x + 10} dx = \int \frac{1}{(x+3)^2 + 1} dx$
 $= \tan^{-1}(x+3) + c$

b) i) $u = x$ $\leftarrow$ $v = \tan^{-1} x$
 $u' = 1$ $\leftarrow$ $v' = \frac{1}{1+x^2}$

$\frac{d}{dx}(x \tan^{-1} x) = \tan^{-1} x + \frac{x}{1+x^2}$

$\therefore \tan^{-1} x = \frac{d}{dx}(x \tan^{-1} x) - \frac{x}{1+x^2}$

$\therefore \int \tan^{-1} x dx = \int \left(\frac{d}{dx}(x \tan^{-1} x) - \frac{x}{1+x^2} \right) dx$
 $= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + c$

c) i) $\frac{dT}{dt} = k(15-T)$ $t=0$ $T=-5$

given $T = 15 - Ae^{-kt}$
 $\frac{dT}{dt} = kAe^{-kt}$ *

since $15-T = Ae^{-kt}$

$\therefore \frac{dT}{dt} = k(15-T)$

ii) $-5 = 15 - Ae^0$
 $A = 20$ $\therefore T = 15 - 20e^{-kt}$

iii) $t=0$ $\frac{dT}{dt} = 5$ sub into *

$5 = k \cdot 20 e^0$
 $\frac{5}{20} = k$ $\therefore k = \frac{1}{4}$

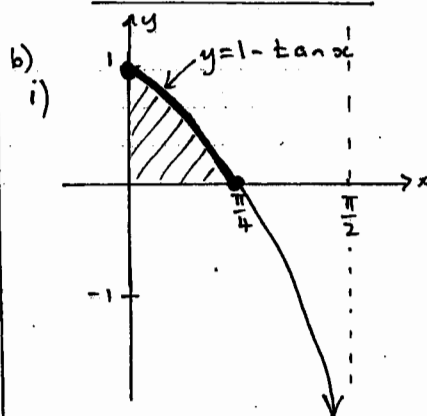
iv) $t=5$ $-\frac{1}{4} \cdot 5$
 $T = 15 - 20e^{-1.25}$
 $T = 9.27^\circ \text{C}$

v) $0 = 15 - 20e^{-\frac{1}{4}t}$
 $20e^{-\frac{1}{4}t} = 15$
 $e^{-\frac{1}{4}t} = \frac{3}{4}$
 $-\frac{t}{4} = \ln \frac{3}{4}$
 $t = 1.1507 \dots \text{min}$
 $\therefore t = 69 \text{ seconds}$

Question 6

a) side x $A = x^2$
 $\frac{dx}{dt} = 0.08$ $\frac{dA}{dx} = 2x$
 $\therefore \frac{dA}{dt} = \frac{dA}{dx} \cdot \frac{dx}{dt}$
 $= 2x \times 0.08$

sub $x = 7$
 $\therefore \frac{dA}{dt} = 1.12 \text{ cm}^2/\text{min}$



ii) $A_x = \int_0^{\pi/4} (1 - \tan x) dx$
 $= \int_0^{\pi/4} 1 - \frac{\sin x}{\cos x} dx$
 $= \left[x + \ln |\cos x| \right]_0^{\pi/4}$
 $= \left(\frac{\pi}{4} + \ln \cos \frac{\pi}{4} \right) - (0 + \ln \cos 0)$
 $= \frac{\pi}{4} + \ln \frac{1}{\sqrt{2}}$
 $= \frac{\pi}{4} + \ln 2^{-1/2}$
 $= \frac{\pi}{4} - \frac{1}{2} \ln 2$
 $= \frac{\pi - \ln 4}{4} \text{ unit}^2$

iii) $V_x = \pi \int_0^{\pi/4} (1 - \tan x)^2 dx$
 $= \pi \int_0^{\pi/4} (1 - 2 \tan x + \tan^2 x) dx$
 $= \pi \int_0^{\pi/4} (\sec^2 x - 2 \tan x) dx$
 $= \pi \left[\tan x + 2 \ln |\cos x| \right]_0^{\pi/4}$
 $= \pi \left[\left(\tan \frac{\pi}{4} + 2 \ln \left(\cos \frac{\pi}{4} \right) \right) - (0 + 2 \ln 1) \right]$
 $= \pi \left[1 + 2 \ln \frac{1}{\sqrt{2}} \right]$

or may tidy up to
 $= \pi \left[1 + 2 \ln 2^{-1/2} \right]$
 $= \pi \left[1 - \ln 2 \right] \text{ units}^3$